

TinyGPT Assignment Report

1. Introduction

This project implements a TinyGPT character-level transformer model trained on the TinyShakespeare dataset. The objective was to build a decoder-only transformer architecture from scratch using PyTorch and analyze training behavior, validation performance, and generated text quality.

2. Model Architecture

The model consists of token embeddings, positional embeddings, 4 GPT transformer blocks, multi-head causal self-attention, feed-forward MLP layers with GELU activation, residual connections, layer normalization, and a final language modeling head. The model contains 145,344 trainable parameters.

3. Hyperparameters

Block Size: 128
Embedding Dimension: 64
Number of Attention Heads: 4
Number of Transformer Layers: 4
MLP Hidden Dimension: 128
Dropout: 0.1
Learning Rate: 0.0003
Batch Size: 64
Epochs: 5
Steps Per Epoch: 200

4. Training Results

The training loss consistently decreased throughout the training process. This shows that the model successfully learned character-level patterns and structures from the TinyShakespeare dataset.

5. Validation Results

The validation loss gradually improved across epochs and reached a final validation loss of 2.7268. This indicates that the model generalized reasonably well without significant overfitting.

6. Generated Text Analysis

Generated text quality improved significantly from checkpoint 1 to checkpoint 5. Earlier checkpoints produced highly random and repetitive outputs, while later checkpoints generated more structured words and sentence-like patterns. The model learned punctuation, spacing, and Shakespeare-style formatting.

7. Temperature Comparison

Temperature 0.5 generated more coherent and readable text with lower randomness. Temperature 1.0 produced balanced creativity and diversity. Temperature 1.5 produced highly random outputs with many nonsensical character combinations. Lower temperature values generate more coherent text, while higher temperatures increase randomness.

8. Challenges Faced

Major challenges included implementing causal masking correctly, handling tensor reshaping for multi-head attention, preventing dimension mismatches, stabilizing training, and implementing autoregressive generation properly.

9. Observations and Conclusion

This assignment demonstrated the working principles of transformer-based language models. The model progressively improved during training and learned meaningful text structures. Temperature sampling had a major impact on output quality and randomness. Overall, the project provided practical experience with GPT-style architectures and attention mechanisms.

Training Loss and Validation Loss

Epoch	Train Loss	Validation Loss	Epoch Time (sec)
1	9.5142	3.1850	192.80
2	3.2858	2.8702	120.91
3	3.0543	2.7804	131.87
4	2.9575	2.7368	151.83
5	2.9108	2.7268	140.90

Generated Text Samples

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